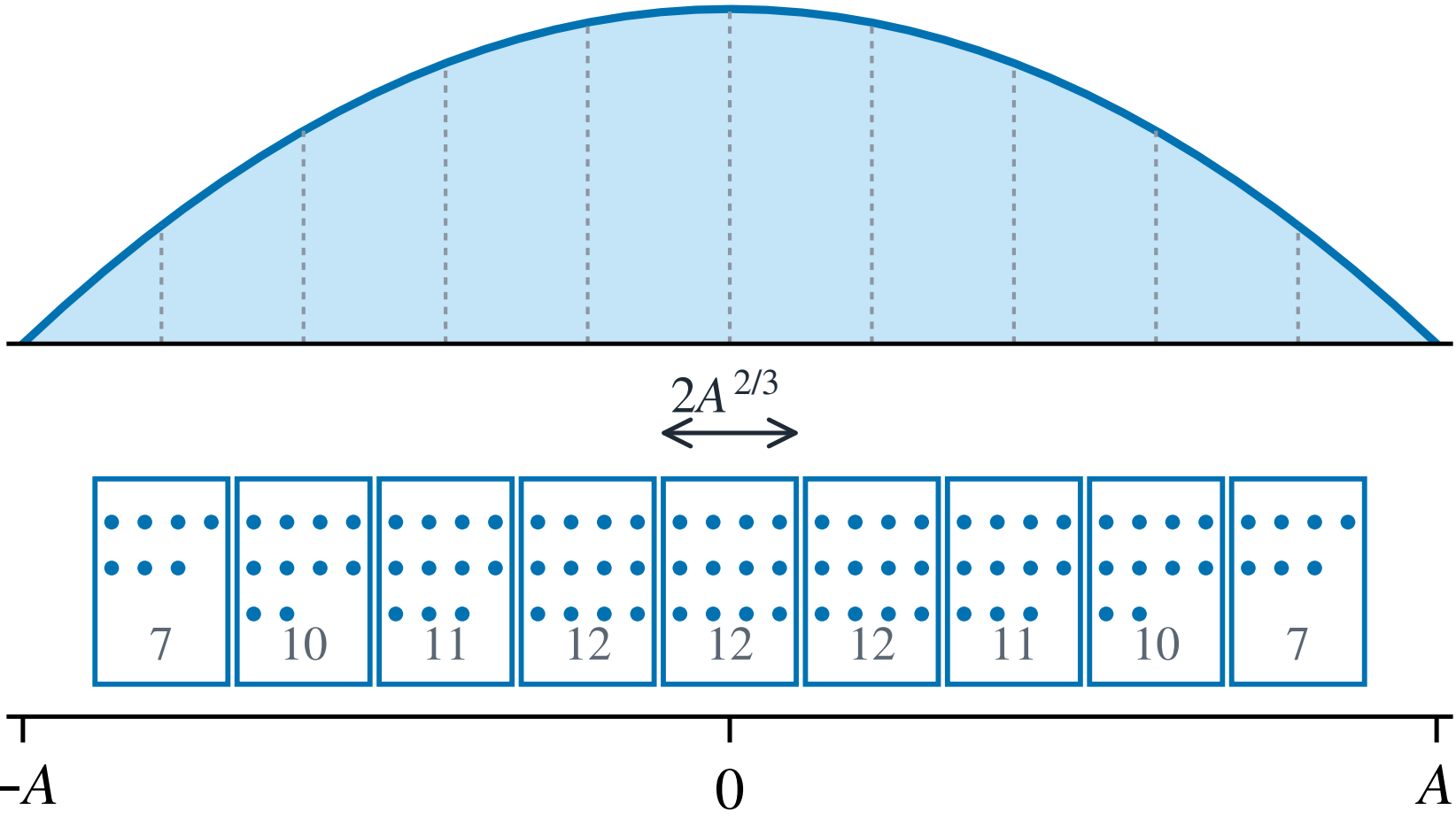
Step 1: p is flat up to $\eta(x)$, with $\log\frac{1}{\eta} \asymp A^{2/3}\left(1-\frac{x^2}{A^2}\right)$



Step 2: each window W_j holds $\gtrsim \frac{l}{\pi}\sqrt{2\log(1/\eta_j)} \asymp A$ points;

about $A^{1/3}$ windows, so $K(A) \gtrsim A \cdot A^{1/3} = A^{4/3}$.